Sindh Science Learning Portal

Class 5 · Science

Chapter 1

Living Things and Classification



Learn how living things are different from non-living things and how we sort them into groups.

Following the Sindh Textbook Board curriculum

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CHAPTER 1

Living Things and Classification

What we will learn Learn how living things are different from non-living things and how we sort them into groups.

1 What is a living thing?

Look around you. You can see **living things** like people, birds, trees and ants. You can also see **non-living things** like stones, books and chairs.

All living things share some features:

- They **grow** bigger with time.
- They **breathe** (take in air).
- They need **food** and **water**.
- They **move** on their own. Plants move slowly too, for example a sunflower turns towards the sun.
- They **reproduce** (make young ones like themselves).
- They **respond** to things around them, like heat, light and touch.

2 Non-living things

Non-living things do not grow, breathe, eat or reproduce on their own.

Some non-living things are **natural**, like stones, water, air and sand. Others are **made by people**, like pencils, bags, cars and fans.

A car moves, but it is not living. It moves only when a person drives it and it has petrol. It cannot grow or make baby cars!

3 Groups of living things: plants and animals

Scientists sort living things into groups. This is called **classification**. It makes them easier to study.

The two biggest groups are **plants** and **animals**.

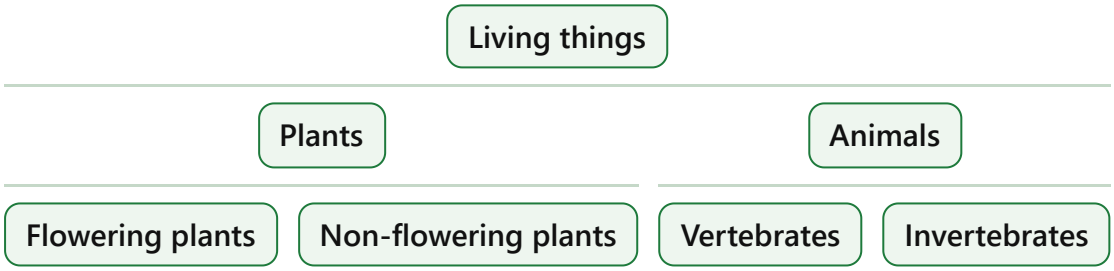
- **Plants** make their own food using sunlight. They cannot walk from place to place.
- **Animals** cannot make their own food. They eat plants or other animals. Most animals can move from place to place.

Plants can be **flowering plants** (like mango, rose and wheat) or **non-flowering plants** (like ferns and mosses).



Picture: Stemonitis · CC BY-SA 2.5 · Wikimedia Commons

Diagram: the parts and what they do



Living things (جاندار · جاندار)
Things that grow, breathe, need food and water, and reproduce.

Plants (پودے · ہوتا)
Living things that make their own food using sunlight.

Animals (جانور · جانور)
Living things that eat plants or other animals. Most can move from place to place.

Flowering plants (گل وارا ہوتا · پھول دار پودے)
Plants that have flowers, fruits and seeds. Examples: mango, rose, wheat.

Non-flowering plants (بنا گلن وارا ہوتا · بغیر پھول والے پودے)
Plants that do not have flowers. Examples: fern, moss.

Vertebrates (ریڑھ دار جانور · فقاریہ)
Animals with a backbone. Examples: fish, frog, lizard, bird, cow.

Invertebrates (بنا ریڑھ وارا جانور · غیر فقاریہ)
Animals without a backbone. Examples: butterfly, snail, spider, ant.

4 Animals with and without a backbone

Animals are sorted by whether they have a **backbone** (a line of bones down the back).

- **Vertebrates** have a backbone. Examples: fish, frog, lizard, sparrow, cow and humans.
- **Invertebrates** do not have a backbone. Examples: butterfly, snail, spider, ant and mosquito.

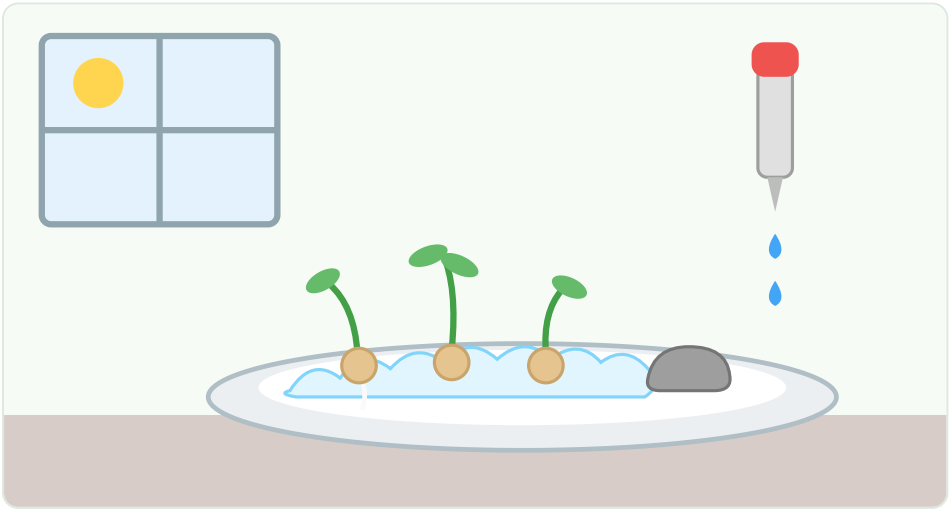
Vertebrates are put into five groups:

- **Fish** live in water and breathe with gills.
- **Amphibians** like frogs live both in water and on land.
- **Reptiles** like lizards and snakes have dry, scaly skin.
- **Birds** have feathers and lay eggs.
- **Mammals** like cows, cats and humans have hair and feed milk to their babies.

Key words

English	سنڌي	اردو
living thing	جاندار	جاندار
non-living thing	غير جاندار	بے جان
classification	درجه بندي	درجہ بندی
reproduce	ٻچا پيدا ڪرڻ	افزائشِ نسل
backbone	ريڑهه جو هڏو	ریڑھ کی ہڈی
vertebrates	ريڑهه دار جانور	فقاریہ
invertebrates	بنا ريڑهه وارا جانور	غیر فقاریہ
amphibians	ٻه جيوت جانور	جل تھلیے
reptiles	رينگڻ وارا جانور	رینگنے والے جانور
mammals	ڪير پيارڻ وارا جانور	ممالیہ

Experiment: Watch a seed come alive



You need

- 4 or 5 chickpea (chana) or moong seeds
- Some cotton wool
- A small plate
- Water
- A small stone

Steps

1. Put the cotton on the plate and make it wet with water.
2. Place the seeds and the small stone on the wet cotton.
3. Keep the plate near a window. Add a few drops of water every day so the cotton stays wet.
4. Look at the plate every day for one week. Draw what you see in your notebook.

Safety: Wash your hands after touching the seeds. Do not eat them.

What did we learn? After a few days the seeds grow small roots and leaves, but the stone does not change. Seeds are alive: they grow when they get water and air. A stone is non-living, so it never grows.

Practice questions

Circle the correct answer

1. **Which of these is a living thing?**

   (a) Stone (b) Chair
   (c) Tree (d) Book

2. **What do living things need to stay alive?**

   (a) Food and water (b) Paint
   (c) Plastic (d) Toys

3. **What do we call it when living things make young ones like themselves?**

   (a) Breathing (b) Reproduction
   (c) Moving (d) Sleeping

4. **Which of these is a non-living thing made by people?**

   (a) Pencil (b) Tree
   (c) Cat (d) Fish

5. **Which of these is a natural non-living thing?**

   (a) Bag (b) Sand
   (c) Fan (d) Car

6. **What is sorting living things into groups called?**

   (a) Classification (b) Breathing
   (c) Digestion (d) Melting

7. **Which living things make their own food?**

(a) Animals (b) Plants

(c) Birds (d) Insects

8. **Which animal has a backbone?**

(a) Butterfly (b) Snail

(c) Fish (d) Spider

9. **Which group of animals has feathers?**

(a) Fish (b) Birds

(c) Reptiles (d) Mammals

10. **A frog lives in water and on land. Which group is it in?**

(a) Amphibians (b) Reptiles

(c) Birds (d) Mammals

Write true or false

11. **A car is a living thing because it moves.** _____

12. **Plants are living things.** _____

13. **An ant does not have a backbone.** _____

14. **Mammals feed milk to their babies.** _____

Match the columns

15. Match each animal with its group.

1. Cow		(a) Bird
2. Sparrow		(b) Reptile
3. Snake		(c) Amphibian
4. Frog		(d) Mammal

Match the columns

16. Match each thing with its type.

1. Stone		(a) Man-made non-living thing
2. Pencil		(b) Plant
3. Mango tree		(c) Animal
4. Goat		(d) Natural non-living thing

Worksheet: Living Things and Classification

Name: _____ Date: _____

A. Fill in the blanks

Word box: classification · feathers · food · vertebrates · water

1. Living things need food and _____.
2. Sorting living things into groups is called _____.
3. Animals with a backbone are called _____.
4. Birds have _____ on their bodies.
5. Plants make their own _____ using sunlight.

B. Put ✓ for true and ✗ for false

1. ☐ A stone grows bigger every year.
2. ☐ Fish breathe with gills.
3. ☐ A snake is a mammal.
4. ☐ Animals cannot make their own food.

C. Match column A with column B (write the number in the brackets)

Column A	Column B
1. Butterfly	( ) Non-flowering plant
2. Rose	( ) Reptile
3. Fern	( ) Invertebrate
4. Lizard	( ) Flowering plant

D. Answer in short

1. Write any three features of living things.

2. A car moves. Why is it not a living thing?

3. Name the five groups of vertebrates.

E. Draw and label

Draw two living things and two non-living things. Write their names.

FOR TEACHERS

Answer key

Chapter 1: Living Things and Classification

Practice questions

1. (c) Tree
2. (a) Food and water
3. (b) Reproduction
4. (a) Pencil
5. (b) Sand
6. (a) Classification
7. (b) Plants
8. (c) Fish
9. (b) Birds
10. (a) Amphibians
11. False
12. True
13. True
14. True
15. 1 → (d), 2 → (a), 3 → (b), 4 → (c)
16. 1 → (d), 2 → (a), 3 → (b), 4 → (c)

Worksheet

A. Fill in the blanks: 1. water · 2. classification · 3. vertebrates · 4. feathers · 5. food

B. Put ✓ for true and ✗ for false: 1. ✗ · 2. ✓ · 3. ✗ · 4. ✓

C. Match column A with column B (write the number in the brackets): (3) Non-flowering plant · (4) Reptile · (1) Invertebrate · (2) Flowering plant

D. Answer in short:

1. They grow, they breathe, and they need food and water.
2. It cannot grow, breathe, eat or reproduce. It moves only when someone drives it.
3. Fish, amphibians, reptiles, birds and mammals.

Picture credits

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